



Unit-01

Introduction to Web Technology & Web Designing



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Outline

- ✓ Introduction to WEB
 - ✓ What is Internet?
 - ✓ What is WWW?
- ✓ How the WEB works?
- ✓ HTTP Protocol
- ✓ Introduction to Web Technologies
 - ✓ HTML
 - ✓ CSS
 - ✓ JavaScript
 - ✓ Bootstrap
 - ✓ Material Design
- ✓ Concepts of Effective Web Design
- ✓ Web Design Issues
- ✓ Planning a website
- ✓ Effective Navigation

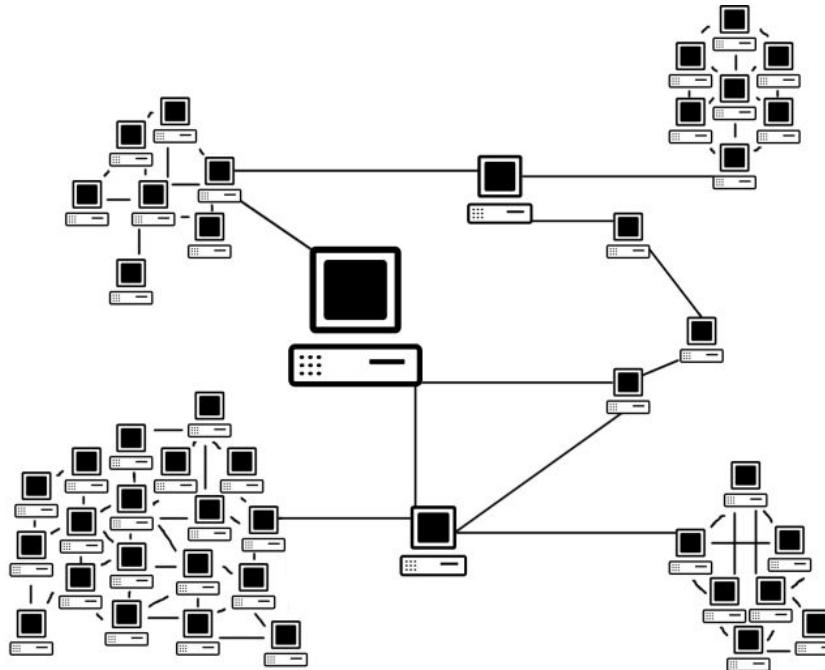


योग: कर्मसु कौशलम्

Darshan
UNIVERSITY

What is Internet?

- ❑ The Internet is a massive **network of networks**, a networking infrastructure.
- ❑ It connects millions of computers together globally, forming a network in which any computer can communicate with any other computer as long as they are both connected to the Internet.
- ❑ Information that travels over the Internet uses many different set of rules which are known as **protocols**.



What is WWW?

- WWW stands for **W**orld **W**ide **W**eb.
- A technical definition of the WWW is – All the resources and users on the Internet that are using HTTP.
- HTTP stands for **H**ypertext **T**ransfer **P**rotocol which is a text-based request-response protocol.
- HTTP is an application layer protocol that allows web-based applications to communicate and exchange the data.
- HTTP is a TCP/IP based protocol, so it also is a connectionless and stateless protocol.
 - After making the request, the client disconnects from the server, then when the response is ready the server re-establish the connection again and deliver the response.
- HTTP is the protocol being used to transfer hypertext documents that makes the World Wide Web possible.

How the Web Works?

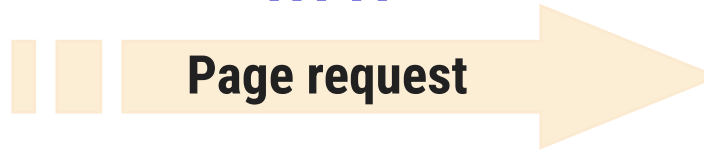
- World Wide Web (WWW) use classical client / server architecture.

Client running a
Web Browser



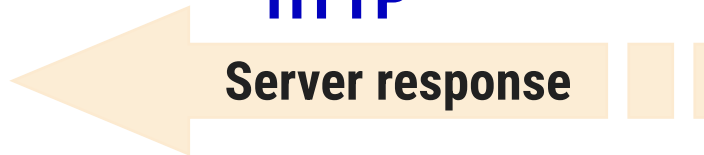
HTTP

Page request



HTTP

Server response



Server running **Web
Server Software**
(Apache, IIS, Tomcat,
etc.)

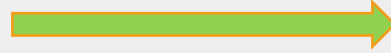


HTTP request

- ❑ The HTTP request message consist of following,
 - ❑ A request line (e.g. GET /index.php HTTP1.1)
 - ❑ Request header fields (e.g. Accept-Language: en)
 - ❑ An empty line (CRLF)
 - ❑ An optional message body
- ❑ The request line and other header fields must end with CRLF (Carriage return, Line Feed) (/r/n)
- ❑ A **request line** contains the **method** of request followed by the **resource** we want and at the end protocol **version** used.
 - ❑ HTTP Request Methods: GET, POST, PUT, DELETE etc...
- ❑ There are many **request header fields** available with HTTP Request, some of are listed below
 - ❑ **Accept** : Media type(s) that is/are acceptable for the response. (e.g. Accept: text/html)
 - ❑ **Accept-Charset**: Character sets that are acceptable. (e.g. Accept-Charset: utf-8)
 - ❑ **Date**: The date and time at which the message was originated (e.g. Date: Tue, 15 Nov 1994 08:12:31 GMT)
 - ❑ **Host**: The domain name of the server (e.g. Host: darshan.ac.in)
 - ❑ **User-Agent**: details of the browser used (e.g. User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/88.0.4324.150 Safari/537.36)

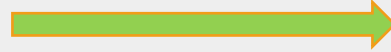
HTTP Request (Example)

GET /index.php HTTP/1.1



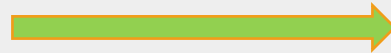
index.php is requested from server using GET method of HTTP version 1.1

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/88.0.4324.150 Safari/537.36



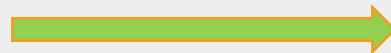
Version 5.0 of mozilla browser is used on windows 10 (64 bit) while requesting the page from the server.

Host: www.darshan.ac.in



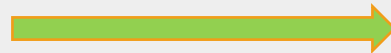
Host of the requested page is www.darshan.ac.in

Accept-Language: en-us



Client accepts US English locale while receiving the response from the server.

Accept: text/html



Client accepts text file containing the HTML in it while receiving the response from the server.

HTTP Response

- After receiving and interpreting a HTTP request message, a server responds with an HTTP response message.
- The HTTP response message consist of following,
 - Status-Line (format= HTTP-Version SP Status-Code SP Reason-Phrase CRLF)
 - *((general-header | response-header | entity-header) CRLF)
 - An empty line (CRLF)
 - An optional message body
- Status-Line consist of
 - HTTP-Version, which can be HTTP/1.1
 - Status-Code is a 3 digit code which is in below format
 - **1xx: Informational** - Request received, continuing process
 - **2xx: Success** - The action was successfully received, understood, and accepted
 - **3xx: Redirection** - Further action must be taken in order to complete the request
 - **4xx: Client Error** - The request contains bad syntax or cannot be fulfilled
 - **5xx: Server Error** - The server failed to fulfill an apparently valid request
 - Reason-Phase is a textual representation of the status code in human readable format.

HTTP Status Codes with reason phrase

"100": Continue	"404": Not Found
"101": Switching Protocols	"405": Method Not Allowed
"200": OK	"406": Not Acceptable
"201": Created	"407": Proxy Authentication Required
"202": Accepted	"408": Request Time-out
"203": Non-Authoritative Information	"409": Conflict
"204": No Content	"410": Gone
"205": Reset Content	"411": Length Required
"206": Partial Content	"412": Precondition Failed
"300": Multiple Choices	"413": Request Entity Too Large
"301": Moved Permanently	"414": Request-URI Too Large
"302": Found	"415": Unsupported Media Type
"303": See Other	"416": Requested range not satisfiable
"304": Not Modified	"417": Expectation Failed
"305": Use Proxy	"500": Internal Server Error
"307": Temporary Redirect	"501": Not Implemented
"400": Bad Request	"502": Bad Gateway
"401": Unauthorized	"503": Service Unavailable
"402": Payment Required	"504": Gateway Time-out
"403": Forbidden	"505": HTTP Version not supported

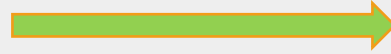


202
Accepted

Refer:
<https://www.w3.org/Protocols/rfc2616/rfc2616-sec6.html> for more details on HTTP Status Code

HTTP Response (Example)

HTTP/1.1 200 OK



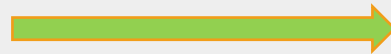
Response is 200 status code with OK message using HTTP1.1

Date: Mon, 27 Jul 2009
12:28:53 GMT



Response Date & Time

Server: Apache/2.2.14 (Win32)



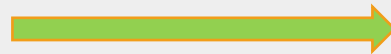
Webserver used by server is Apache and version is 2.2.14 built for 32bit OS

Last-Modified: Wed,
22 Jul 2009 19:15:56 GMT



Last modified at Date & Time.

Content-Length: 88



Content size of the response is 88 bytes

Content-Type: text/html



Content type of the response is text file containing HTML

Introduction to Web Technologies

□ HTML

- HTML stands for Hypertext Markup Language.
- It is used to design the front-end portion of web pages using a markup language.
- HTML is the combination of Hypertext and Markup language. Hypertext defines the link between the web pages. The markup language is used to define the text documentation within the tag which defines the structure of web pages.

□ CSS

- Cascading Style Sheets fondly referred to as CSS is a simply designed language intended to simplify the process of making web pages presentable.
- CSS allows you to apply styles to web pages.
- More importantly, CSS enables you to do this independent of the HTML that makes up each web page.

□ JavaScript

- JavaScript is a famous scripting language used to create magic on the sites to make the site interactive for the user.
- It is used to enhancing the functionality of a website to running cool games and web-based software.

Introduction to Web Technologies (Cont.)

□ Bootstrap

- Bootstrap is the most popular HTML, CSS, and JavaScript framework for developing responsive, mobile-first websites.
- Bootstrap is completely free to download and use! The primary purpose of adding it to a web project is to apply Bootstrap's choices of color, size, font and layout to that project.
- As such, the primary factor is whether the developers in charge find those choices to their liking.
- Once added to a project, Bootstrap provides basic style definitions for all HTML elements.

□ Material Design

- Material is a design system created by Google to help teams build high-quality digital experiences for Android, iOS, Flutter, and the web.
- Material Design is inspired by the physical world and its textures, including how they reflect light and cast shadows.

Concept of Effective Web Design

- ❑ It's a good idea to first think about and design your site. That way, you'll give yourself direction and you'll need to reorganize less later.
- ❑ To design your site:
 1. Figure out why you're creating this site. What do you want to convey?
 2. Think about your audience. How can you tailor your content to appeal to this audience? For example, should you add lots of graphics or is it more important that your page download quickly?
 3. How many pages will you need? What sort of structure would you like it to have? Do you want visitors to go through your site in a particular direction, or do you want to make it easy for them to explore in any direction?
 4. Sketch out your site on paper.
- ❑ Devise a simple, consistent naming system for your pages, images and other external files.

Web Design Issues

- A. Browser & Operating Systems
- B. Bandwidth and Cache
- C. Display Resolution
- D. Look & Feel
- E. Page Layout and Linking
- F. Locating Information
- G. Making Design user-Centric
- H. Sitemap

a) Browser & Operating Systems

- ❑ Web pages are written using different HTML tags and viewed in browser window.
- ❑ The different browsers and their versions greatly affect the way a page is rendered, as different browsers sometimes interpret same HTML tag in a different way.
- ❑ Different versions of HTML also support different sets of tags.
- ❑ The support for different tags also varies across the different browsers and their versions.
- ❑ Same browser may work slightly different on different operating system and hardware platform.
- ❑ To make a web page portable, test it on different browsers on different operating systems.

b) Bandwidth and Cache

- ❑ Users have different connection speed, i.e. bandwidth, to access the Web sites.
- ❑ Connection speed plays an important role in designing web pages, if user has low bandwidth connection and a web page contains too many images, it takes more time to download.
- ❑ Generally, users have no patience to wait for longer time than 10-15 seconds and move to other site without looking at contents of your web page.
- ❑ Browser provides temporary memory called cache to store the graphics.
- ❑ When user gives the URL of the web page for the first time, HTML file together with all the graphics files referred in a page is downloaded and displayed.

c) Display Resolution

- Display resolution is another important factor affecting the Web page design, as we do not have any control on display resolution of the monitors on which user views our pages.
- Display or screen resolution is measured in terms of pixels and common resolutions are 800 X 600 and 1024 X 786.
- We have three choices for Web page design.
 - Design a web page with fixed resolution.
 - Make a flexible design using HTML table to fit into different resolution.
 - If the page is displayed on a monitor with a higher resolution, the page is displayed on left hand side and some part on the right-hand side remains blank. We can use centered design to display page properly.
 - Ideally we should use some frameworks for designing like Bootstrap/Material design.

d) Look & Feel & e) Page Layout and Linking

□ Look & Feel

- Look and feel of the website decides the overall appearance of the website.
- It includes all the design aspects such as
 - Web site theme
 - Web typography
 - Graphics
 - Visual structure
 - Navigation etc...

□ Page Layout and Linking

- Website contains of individual web pages that are linked together using various navigational links.
- Page layout defines the visual structure of the page and divides the page area into different parts to present the information of varying importance.
- Page layout allows the designer to distribute the contents on a page such that visitor can view it easily and find necessary details.

f) Locating Information & g) Making Design User-Centric

□ Locating Information

- Webpage is viewed on a computer screen and the screen can be divided into five major areas such as center, top, right, bottom and left in this particular order.
- The first major area of importance in terms of users viewing pattern is the center, then top, right, bottom and left in this particular order

□ Making Design User-Centric

- It is very difficult for any Web designer to predict the exact behavior of the Web site users.
- However, idea of general behavior of common user helps in making design of the Web site user centric.
- Users either scan the information on the web page to find the section of their interest or read the information to get details.

h) Sitemap

- ❑ A Sitemap is a model of a website's content designed to help both users and search engines navigate the site.
- ❑ Many a times Web sites are too complex as there are a large number of sections and each section contains many pages.
- ❑ It becomes difficult for visitors to quickly move from one part to other.
- ❑ Once the user selects a particular section and pages in that section, user gets confused about where he/she is and where to go from there.
- ❑ To make it simple, keep your hierarchy of information to few levels or provide the navigation bar on each page to jump directly to a particular section.

Planning a Website

- The most important activity in a website development is planning.
- To achieve higher success of the website in terms of user satisfaction, better planning is needed.
- Before we start developing a website, we should ask question such as,
 - Why are we developing this website?
 - What do we achieve by developing this website?
 - Who are the people who will use this website?
 - What are the information contents?
 - How are these contents organized? What are the possible ways?

Effective Navigation

- The most important design element in the web design after page layout is navigation design.
- Navigation means the ways to move from one page to another page in a Web site using hyperlinks provided on the page.
- If navigation design is not proper then user feels the problem in moving around the pages in your site in a desired manner or gets confused and leaves the site.
- Tips for Effective Navigation:
 - Navigation links are either **text based**, i.e. a word or a phrase, or **graphical**, i.e. a image.
 - Navigation links should be **clear** and **meaningful**.
 - It should be **consistent**.
 - Link should be **understandable**.
 - Organize the links such that contents are **grouped logically**.
 - Provide **search** link, if necessary, usually on top of the page.
 - Use **common links** such as 'about us' or 'Contact us'.
 - Provide the way to **return to first page**.
 - Provide the user with **information** regarding **location**.
 - **Horizontal navigation bar** can be provided on each page to directly jump to any section.



***Thank
You***



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